Method Statement

Ref. #: DCP03/08-0207-B-2026



DonProof PU-C100

(Cold applied pure polyurea membrane for waterproofing and protection)

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Section A: General Comments

General Notes:

The information below is a detailed overview of the application of DCP's **DonProof PU-C100** waterproofing and protection system and should be read in conjunction with the relevant technical data sheet prior to application. All DCP Products should be applied by experienced specialist contractors.

All the points below assume the correct preparation of the relevant surface.

High-Temperature Working:

Application temperature ranges from 5°C to 35°C and relative humidity must not exceed 85%. The substrate's temperature must be at least 3°C above the measured due point temperature if any.

It is suggested that, for temperatures above 35°C, the following guidelines are adopted as good working practice:

i. Unmixed materials and equipment should be stored in a cool place and away from direct sunlight.
ii. Avoid application during the peak temperature of the day.
iii. Ensure proper and adequate ventilation.
iv. Plan for enough materials, tools, and labor to ensure a continuous applicant process.

Low-Temperature Working:

It is suggested that, for temperatures below 10°C, the following guidelines are adopted as good working practice:

i. Unmixed materials should be stored in a warm.
ii. Cold temperatures will affect the properties of the material.
iii. Avoid applying the material if the temperature is around 5°C and falling.
iv. Do not apply under rain or snow, and avoid dew points conditions during application.

System Products:

Primer: **DonProof Primer PU, DonProof PM** [if required]

Waterproofing Membrane: **DonProof PU-C100**

Tools and Equipment:

It is suggested that the following list of equipment are adopted as a minimum requirement

Personal protection	:	*Protective overalls*
	:	*Goggles or a face mask*
	:	*Good quality gloves*
	:	*Safety shoes*
	:	*Safety helmet*
Preparation equipment	:	*Concrete vacuum (Fig.1)*
	:	*Grit blasting machine (Fig.2)*
	:	*Stiff wire brush (Fig.3)*
	:	*Grinding machine (Fig.4)*
Mixing equipment	:	*Power-whisk fitted in a heavy-duty slow speed electric drill (Fig.5)*
	:	*Squeegee (Fig.6)*
Application equipment	:	*Lamb's wool roller (Fig.7)*
	:	*Brush (Fig.8)*





Fig.1: Concrete vacuum



Fig.2: Grit blasting machine



Fig.3: Stiff wire brush



Fig.4: Grinding machine



Fig.5: Power-whisk fitted in a heavy-duty slow speed electric drill



Fig.6: Squeegee



Fig.7: Lamb's wool roller



Fig.8: Brush

Section B: Application

1.0 Substrate Preparation



1.1 Concrete Substrates

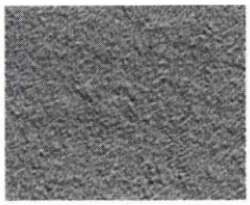
1.1.1 New concrete or cementitious substrates should be at least 28 days old. Maximum relative humidity should not exceed 90%. Perform a relative humidity test using in situ devices according to ASTM F2170.

1.1.2 The substrate (new or existing) should be dry, smooth, clean, sound, and free from any contamination such as mortar and paint splashes, curing compounds, silicones, oil, and grease.

1.1.3 The surface must have a minimum compression strength of 25 MPa and a minimum adhesion strength of 1.5 MPa.

1.1.4 Excess laitance deposits are best removed by grit blasting or wire brushing followed by vacuum cleaning to remove dust debris. All preparation equipment should be of a type approved by DCP.

1.1.5 Concrete surface should achieve a minimum surface roughness of (CSP 3 to 6), in accordance with the following ICRI recommendations:



CSP 1
(acid etched)



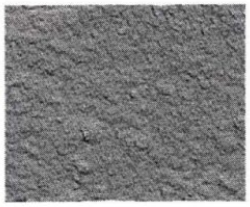
CSP 2
(grinding)



CSP 3
(light shotblast)



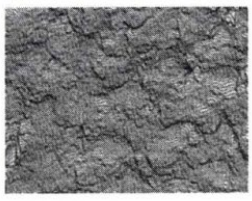
CSP 4
(light scarification)



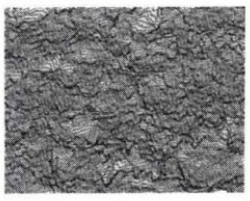
CSP 5
(medium shotblast)



CSP 6
(medium scarification)



CSP 7
(heavy abrasive blast)



CSP 8
(scabbled)



CSP 9
(heavy scarification)

1.1.6 The processes to eliminate the laitance on the concrete and to open the pore of it, are:

- **Grinding/Milling:** Using a rotary machine with specialty grinding wheels with tungsten carbide; this method is suitable for removing concrete paste and other hard substances. It causes surface erosion by rubbing with stones or hard sanding discs. This removes the softest parts of the surface, for example, surface laitance, which sometimes occurs during concrete pouring or curing. But is not suitable for soft existing coatings or when the friction caused by heat meets this product.



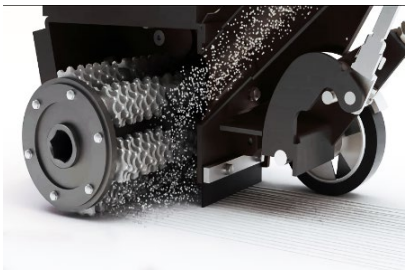
- **Shot blasting:** involves the projection or direct impact of steel grit of different sizes on the concrete's surface. This process produces very little dust. Some water cleaning or vacuuming may be required after the use of this method. This is effective for the removal of concrete surface paste, sealers, and hard coatings. Speed and number of passes over an area determine the depth of removal. Edging and small spaces are not susceptible to this type of removal without small specially designed equipment.



- **Sandblasting:** It uses sand or a type of abrasive shot by compressed air through a nozzle. Sandblasting is recommended for horizontal, vertical, and overhead use. This method is recommended for the removal of the surface of concrete, existing sealers, and hard coatings. The creation of dust may be prohibited by environmental regulation. Vacuum mechanisms are available to remove dust from the air. Wet sandblasting is available that complies with environmental regulations. This method of surface preparation, while highly effective, has lost utility where environmental regulation has restricted its use.



- **Scarifying:** Generally, employs a rotary machine. This method utilizes specialty blades or impact devices to break the surface of concrete or coatings. Several passes over the same area may be required to remove the existing surface completely. In the case of elastomeric surface coatings, blade shaving may be the only effective method of removal where water use is a problem. Most sealers and coatings will not allow removal by acid etching. Most elastomeric coatings tend to "bounce" shot blasting.



Notes:

- *Applicator must decide in each case the most suitable method, depending on the conditions of the substrate or surface or the desired result (always in combination with the primer to be used).*
- *To complete these processes prior to application, the substrate must be cleaned using aspiration equipment, thus avoiding the need to use water as this could hamper adherence of the membrane.*

1.1.7 Surface defects and imperfections such as voids, small cracks, and blowholes should be repaired before application using **Quickmast 341** epoxy paste to prevent material flowing into them.



1.1.8 Substrate should be free of any standing water and does not suffer from conditions of rising damp, as this will negatively affect the membrane's adhesion.

1.1.9 If the substrate is known to suffer from rising damp, a damp-proof membrane should be used (**DonProof DPM**).

1.2 Metal Substrates

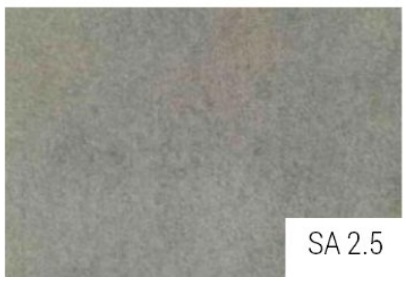
1.2.1 Most metallic surfaces will require blast cleaning using either a dry blast of silica sand or water/wet blasting.



1.2.2 Subsequent cleaning of these metals should be carried out using pressurized water or acetone-based solvents to simultaneously clean and remove any grease from the surface.

1.2.3 In many cases, the surface passivation is required before application of the products.

1.2.4 All steel surfaces should be grit blasted to reach a bright, near white finish meeting the requirement of Swedish Standard SA 2 1/2 or according to the SSPC- SP10.



1.2.5 After the abrasion process, and before the waterproofing system application, is necessary to apply corrosion inhibitor products.

1.3 Joints and moving cracks:

1.3.1 **DonProof PU-P60** shouldn't be installed over any non-filled/sealed joints or moving cracks.

1.3.2 Open up and clean the existing joints in between the concrete slab and vacuum thoroughly**.**

1.3.3 All dust, loose and friable material must be removed from all joint voids before application of any joint sealant.

1.3.4 All existing joints such as (expansion, isolation, construction, and control joints) as well as all moving cracks, must be sealed using **Flexseal PU425/PU440**.

2.0 Priming

Concrete Surfaces:

DonProof PU-C100 is designed to be used with **DonProof Primer PU** whereas for surfaces with a relative humidity greater than 75% **DonProof DPM** is recommended.

Immediately prior to priming, the substrate should be thoroughly cleaned to remove any remaining traces of dust or other loose material.

2.1 **DonProof Primer PU** [For application onto concrete substrates]

2.1.1 **DonProof Primer PU** is a low viscosity polyurethane-based primer that is designed to enhance the adhesion of waterproofing membrane with porous concrete substrates.

2.1.2 **DonProof Primer PU** will penetrate, stabilize and seal the concrete surface in depth. Porous substrates may require a second primer coat – when the first coat is directly absorbed into the substrate – but minimum over coating times must still be observed.

2.1.3 Apply **DonProof Primer PU** at a rate of (0.1 - 0.2) ltr/m^2, depending on the substrate porosity, using brush or rollers.

Note: Avoid any primer ponding on the floor.



2.1.4 If a second coat is required, apply a second coat of the primer within its overcoating time.

2.1.5 Allow to dry fully for 24 hours before applying **DonProof PU-C100.**

2.1.6 The primer should be protected from damp, condensation, and water for at least 24 hours.

2.2 **DonProof DPM** [For application onto surfaces with high relative humidity/rising damp]

2.2.1 Stir individual components of **DonProof DPM** and ensure that the bottom and sides are thoroughly scraped.

2.2.2 Transfer the entire content of the hardener into the base and mix for 2 - 3 minutes using a slow-speed mixer fitted with a suitable paddle.

2.2.3 Use brush or short hair lamb's wool roller for application to the prepared surface.

2.2.4 Prime with 1 - 2 coats of **DonProof DPM** depending on surface relative humidity at a rate of (5 m^2/kg per coat) to achieve a dry film thickness of 200 microns per coat and allow to dry.

2.2.5 Allow to dry fully for 24 hours before applying **DonProof PU-C100**.

2.3 **Non-porous Surfaces**

2.3.1 Steel, Aluminum, ceramic and porcelain tiles, should be primed using **DonProof Primer PU** as described above.

2.3.2 Consult DCP Technical Department for specific primer recommendation based on the project conditions.

*Note: It is recommended to test the adhesion between the substrates and **DonProof PU-C100** on small area onsite prior to proceeding with the works in order to ensure that the substrate is well prepared and compatible with the product and primer.*

3.0 Mixing

3.1 Stir **DonProof PU-C100** individual components well before use to homogenize the pigment and disperse any settlement using a low-speed mixer. Failing to do so may cause color variances, foaming, sticky coating, and negatively affect performance.

3.2 Taking care to ensure that the bottom and sides are thoroughly scraped. Transfer the entire contents of component B into component A container and mix slowly for at least 3 minutes until a homogeneous mix is achieved.

4.0 Application

4.1 Use brush, squeegee, lamb's wool roller, or airless spray machine to apply the mixed **DonProof PU-C100** onto the prepared and primed surface.



4.2 If needed, a second coat should be applied at a right angle to the first coat after 4 hours to 7 days.

4.3 Duct tape or masking tape can be used, but must be removed immediately after spraying whilst the Polyurea is still wet to form a defining straight edge.

4.4 Application of subsequent layers should be within the overcoating open window time. If maximum overcoating time is exceeded, it is necessary to activate the previous layer using a suitable primer.



4.5 Theoretical consumption is approximately:1.8 – 2.0 kg / m² to provide 1.5 mm dry film thickness.

4.6 A visual inspection shall take place for any pinholes and or defects that will require repair.

5.0 Protection Coat Application

5.1 As with all aromatic coatings, **DonProof PU-C100** should be sealed with a suitable polyurethane protection coat to prevent yellowing due to UV rays exposure. However, this will not cause any negative effect on the physical properties of the product.

5.2 Consult DCP Technical Department for specific recommendations to guarantee the durability.

6.0 Cleaning

6.1 Tools and equipment can be cleaned with DCP Solvent PU.

7.0 Limitations

7.1 Special care should be taken to provide an unbroken coating at external corners and similarly exposed protrusions.

7.2 **DonProof PU-C100** doesn't require any special curing but must be protected from rain, and water until the coating has been cured.

7.3 Application should not be carried out when there is standing or running water.

7.4 When exposed to UV, a slight discoloration of the membrane is probable. This will not affect, in any way, the waterproofing properties of the final coat.

7.5 At low temperatures extra care is required when storing and mixing **DonProof PU-C100**, before mixing and processing ensure that the product is stored at temperature no less than 15°C, low temperatures result in high viscosity (thicker liquid) and will lead to poor mixing and failure in application delivery.

Section C: Cautions

Health and safety

- Apply in well-ventilated areas. In closed areas use force ventilation and wear approved respirators when necessary.
- **DonProof PU-C100** should not come into contact with skin and eyes. In case of contact with eyes wash immediately with plenty of water and seek medical advice promptly.

Fire:

DonProof PU-C100 is nonflammable. Do not use near-high heat or open flame.

For further information on refer to the Material Safety Data Sheet.

Section D: Approval and Variations

This method statement is offered by DCP as a 'standard proposal' for the application of **DonProof PU-C100**. It remains the responsibility of the Engineer to determine the correct method for any given application. Where alternative methods are to be used, these must be submitted to DCP for approval, in writing, prior to commencement of any work. DCP will not accept responsibility or liability for variations to the above method statement under any other condition.